

DIDAR ALI

AI/ML Engineer | Computer Systems Engineer | Researcher

didarali1129@gmail.com | +923420032958 | Islamabad, Pakistan

github.com/didar-ali-deed | linkedin.com/in/didar-ali-deed | didar-portfolio-web.vercel.app

PROFESSIONAL SUMMARY

Results-driven AI/ML Engineer and recent B.Sc. Computer Systems Engineering graduate (UET Peshawar, 2025) with hands-on experience building end-to-end deep learning pipelines, computer vision systems, and NLP solutions. Published peer-reviewed researcher (Wav2Vec2 transformer achieving 79% accuracy on noisy speech emotion datasets). Proficient in Python, PyTorch, TensorFlow, FastAPI, and production model deployment using onnxruntime and Docker. Passionate about applying machine learning and AI to solve real-world problems in identity verification, document intelligence, health diagnostics, and human-computer interaction.

TECHNICAL SKILLS

- **AI & Machine Learning:** PyTorch, TensorFlow, Scikit-learn, XGBoost, Wav2Vec2, BERT, LSTM, CNN, Transformers, Transfer Learning, Fine-Tuning
- **Programming Languages:** Python, C, C++, SQL, Bash/Shell Scripting
- **Data Science & NLP:** Pandas, NumPy, Matplotlib, Seaborn, Librosa, EDA, HuggingFace Datasets
- **MLOps & Deployment:** FastAPI, Flask, Streamlit, Docker, onnxruntime, REST APIs, AWS (Basic)
- **Systems & Tools:** Linux/Bash, Git, GitHub, SQLite, VS Code, OS concepts, Memory Management
- **Soft Skills:** Analytical Thinking, Research Mindset, Problem Solving, Team Collaboration, Fast Learner

EXPERIENCE

AI/ML Engineer | Self-Employed (Freelance)

Jan 2026 – Present, Remote

- Designing and deploying production-grade computer vision systems, including face recognition with anti-spoofing liveness detection
- Integrating trained ML models into scalable REST APIs (FastAPI, Flask) for real-time AI inference
- Containerizing AI applications with Docker for portable, reproducible production deployments
- Focusing on identity verification and document intelligence use cases

Final Year Researcher | UET Peshawar; FYP

2024 – 2025, Peshawar, Pakistan

- Designed a Wav2Vec2 transformer-based Speech Emotion Recognition system achieving 79% accuracy on noisy real-world datasets
- Built complete ML pipeline: data preprocessing, noise augmentation (RAVDESS/TESS), feature extraction, model training, and evaluation
- Research published in VFAST Transactions on Software Engineering
- Demonstrated reproducible experimental workflows and thorough model benchmarking

PROJECTS

Speech Emotion Recognition System (FYP); *Python, PyTorch, Wav2Vec2, Librosa, Transformers*

- Achieved 79% classification accuracy on noisy speech datasets using a Wav2Vec2 pre-trained transformer model
- Built end-to-end pipeline: audio preprocessing, noise augmentation, feature extraction, model fine-tuning, and evaluation
- Published findings in a peer-reviewed journal; VFAST Transactions on Software Engineering (2025)
- GitHub: github.com/didar-ali-deed/Vocal-Sentiments-Transformer-Based-SER

BERT Sentiment Analysis; FastAPI + React; *Python, PyTorch, BERT, FastAPI, React*

- Fine-tuned BERT on the IMDb dataset for binary sentiment classification (Positive / Negative)
- Built FastAPI async inference backend with real-time prediction endpoints and interactive React frontend
- GitHub: github.com/didar-ali-deed/sentiment-analysis-bert-fastapi-imdb

AI Health Super App; *Python, TensorFlow, XGBoost, CNN, DNN, Streamlit, SQLite*

- Built multi-disease AI diagnostic platform predicting Diabetes, Parkinson's Disease, and Pneumonia
- Implemented multi-model ensemble (CNN, DNN, XGBoost) for robust and reliable predictions
- Designed end-to-end pipeline from data preprocessing to real-time inference with Streamlit dashboard and SQLite logging
- GitHub: github.com/didar-ali-deed/health-ai-super-app

Virtual File System (Linux); *C, OS Concepts, File Systems, Linux*

- Implemented a Linux-style virtual file system from scratch in C with full permission control, directory hierarchy, and file metadata

Emotion Detection; LSTM; *Python, PyTorch, LSTM, HuggingFace*

- Text-based multi-class emotion classification using LSTM networks trained on HuggingFace datasets

PUBLICATION

Vocal Sentiments: Transformer-Based Speech Emotion Recognition

VFAST Transactions on Software Engineering, 2025

vfast.org/journals/index.php/VTSE/article/view/2174

EDUCATION

B.Sc. Computer Systems Engineering

2021 – 2025

University of Engineering & Technology (UET), Peshawar

Higher Secondary School Certificate (Pre-Engineering)

2019 – 2021

Government Degree College, Chitral

Secondary School Certificate (Science)

2017 – 2019

Al Karim Public School Susoom, Chitral

CERTIFICATIONS

- Google Data Analytics Professional Certificate; Google / Coursera
- Google Cybersecurity Professional Certificate; Google / Coursera
- Programming for Everybody (Python); University of Michigan / Coursera

References available upon request, Open to remote and on-site opportunities